

Unit 7, Lesson 1: The Pythagorean Theorem



Benchmarks

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



Learning Target

- ☐ I can find the missing side of a right triangle using the Pythagorean Theorem.



7.1.1 Warm-Up: Sales Strategist

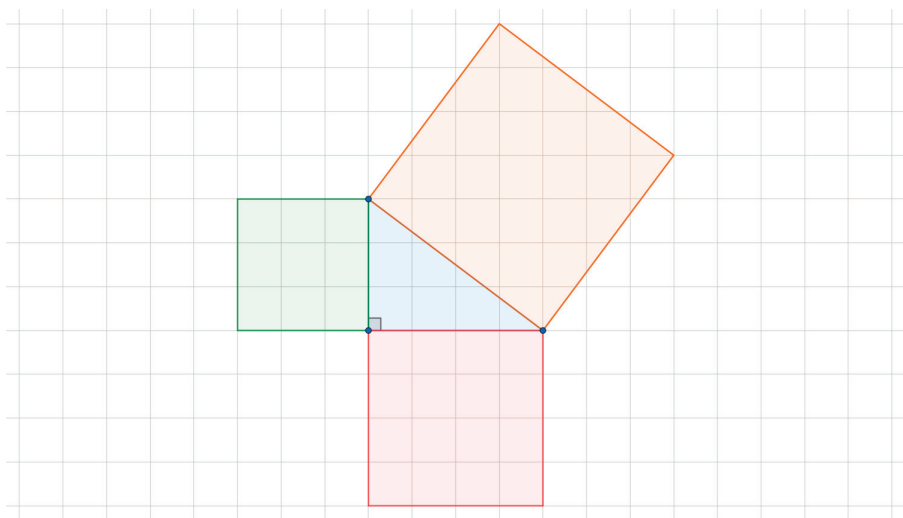
Kishan Patel says that he was never really a good student in high school. He graduated with a 2.9 GPA. All of those challenges prepared him for business. Business is about failing and doing better. Every time you fail, you have to improve in some way to get to the next level.

1. Based on the statement above, how do you think Kishan views failure?
2. When you fail at something, describe how you typically react.

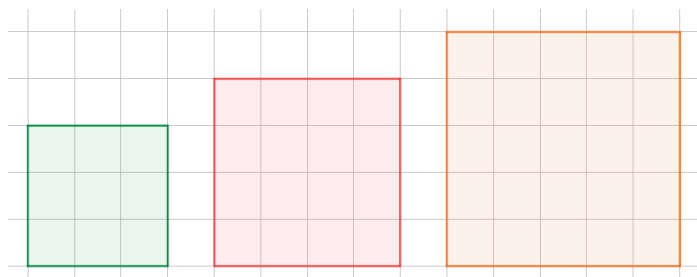


7.1.2 Collaboration: Pythagorean Theorem

A right triangle is shown with three squares.

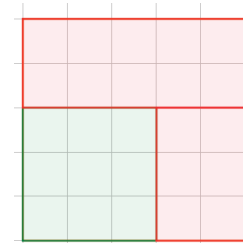


1. The three squares from the diagram are shown. Complete the table with the area of each square.



Side Length (units)	Area of Square (square units)
3	
4	
5	

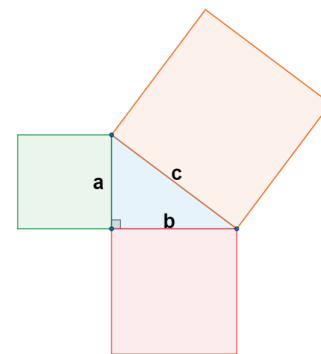
2. Explain how the diagram shows that the area of the green square plus the area of the red square is equal to the area of the orange square.



3. Complete the relationship of the areas by filling in the box.

$$3^2 + 4^2 \square 5^2$$

4. Use the diagram to write an explanation of the Pythagorean Theorem, $a^2 + b^2 = c^2$.



5. Iverson and Jake are at the beach and plan to meet a friend for lunch.



Iverson decided to walk directly to the restaurant, while Jake decides to take a route that will allow him to cross the street using the crosswalk. The diagram shows the two routes. Jake walks horizontally 252 meters (m) and then vertically 115 m.



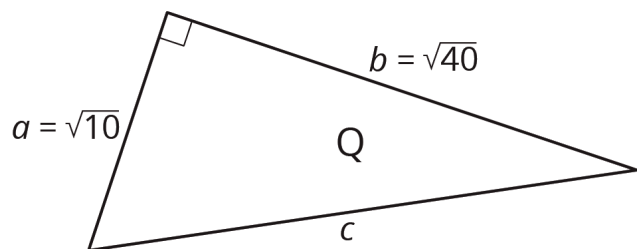
- a. Complete the table.

	Iverson	Jake
Distance Walked		

- b. Discuss with your partner the advantages and disadvantages of each person's walk to the restaurant. Summarize your discussion in the table.

Person	Advantages	Disadvantages
Iverson		
Jake		

6. A right triangle is shown with side lengths $\sqrt{10}$ and $\sqrt{40}$ units.



Find the length of the hypotenuse using the Pythagorean Theorem. If the measure of the side length is an irrational value, write an exact solution and an approximate solution.

- a. State Pythagorean Theorem

$$a^2 + b^2 = c^2$$

- b. Substitute

$$\underline{\hspace{1cm}}^2 + \underline{\hspace{1cm}}^2 = c^2$$

- c. Solve for the missing variable.

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = c^2$$

$$\underline{\hspace{1cm}} = c^2$$

- d. Solutions

Exact solution:

$$c = \sqrt{\hspace{1cm}}$$

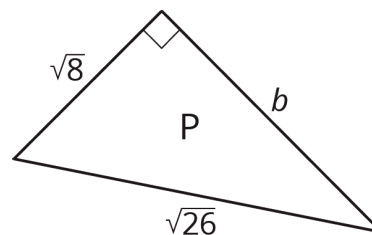
Approximate answer:

$$c \approx \underline{\hspace{1cm}}$$

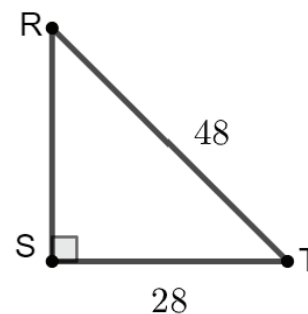


7.1.3 Guided Practice: Pythagorean Theorem

1. Find the exact measure of b in the diagram.

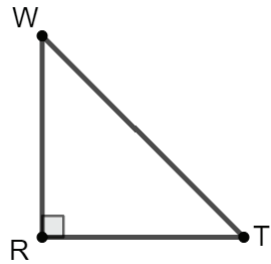


2. $\triangle RST$ is shown, where $ST = 28$ and $RT = 48$.



Find the length of $\overline{SR}$ to the nearest thousandth.

3. $\triangle WRT$ is shown, where $WT = 18.4$ and $RT = 12.1$.



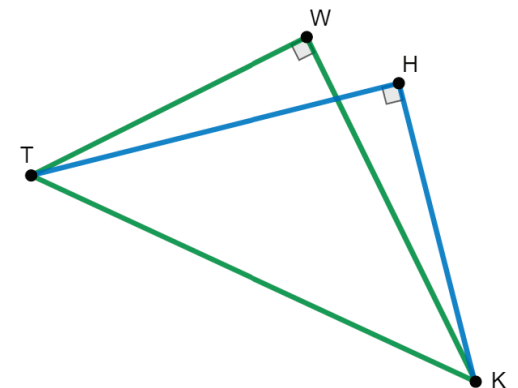
Find the length, to the nearest thousandth, of $\overline{WR}$.



7.1.4 Your Turn!

1. In $\triangle HMX$, $m\angle MXH = 90^\circ$, $MH = 12$, and $MX = 7$. Determine the length of $\overline{XH}$. Round to the nearest thousandth.

2. $\triangle TWK$ and $\triangle THK$ are shown, where $TW = \sqrt{45}$, $TH = \sqrt{68}$, and $TK = \sqrt{113}$. Find the length of $\overline{WK}$ and $\overline{HK}$. Round to the nearest thousandth.





7.1.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about the Pythagorean Theorem.

Unit 7, Lesson 2: Defining Sine and Cosine



Benchmarks

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



Learning Target

- ☐ I can define sine and cosine using similar triangles.